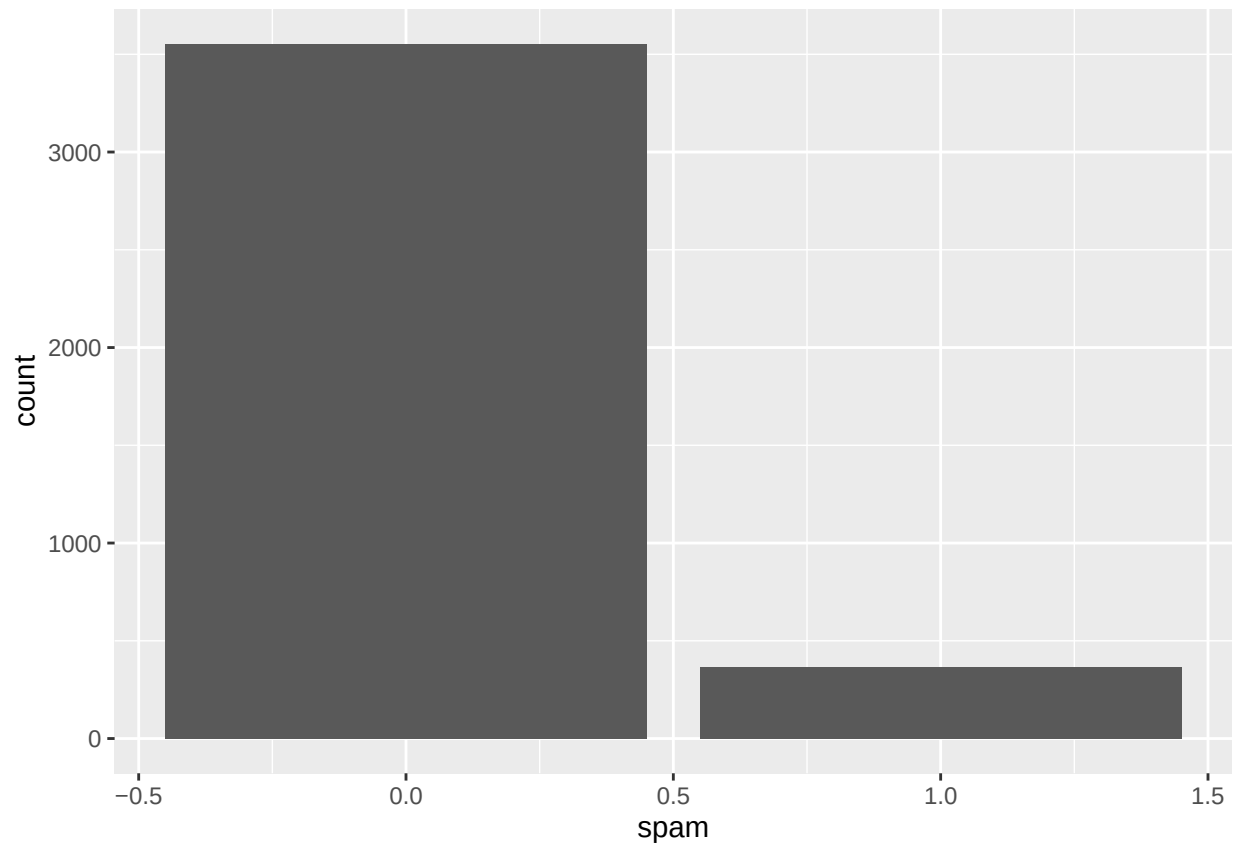


```
library(ggplot2)

email <- read.csv("https://raw.githubusercontent.com/roualdes/data/master/email.csv")

ggplot(data = email, aes(spam)) +
  geom_bar()
```



Two Sided

```
phat <- mean(email$spam)
n <- length(email$spam)
s <- sd(email$spam)
alpha <- 0.02 # two sided

se <- s / sqrt(n)
t.star <- qt(1 - alpha / 2, df = n - 1)
me <- se * t.star
c(phat - me, phat + me)
```

```
## [1] 0.08277167 0.10442547
```

One Sided Upper

```
phat <- mean(email$spam)
n <- length(email$spam)
s <- sd(email$spam)
```

```
alpha <- 0.02 # two sided  
se <- s / sqrt(n)  
t.star <- qt(1 - alpha, df = n - 1)  
me <- se * t.star  
c(phat - me, Inf)
```

```
## [1] 0.08404108      Inf
```

Mean Chars per Email

```
mean(email$num_char)
```

```
## [1] 10.70659
```